Robot Design Talk: Team Script

Rosie Robotics, Team #75872 · Robot: Rosie Rover

This copy has **everything on the judges' handout**, plus the rubric and **what each person says**. The judges' handout has the same pictures and tables, so when you point at something here, the judges can find it on their copy.

We have **5 minutes**. We go in the same order as the judges' Robot Design rubric: **Identify → Design → Create → Iterate → Communicate**.

Tips for everyone

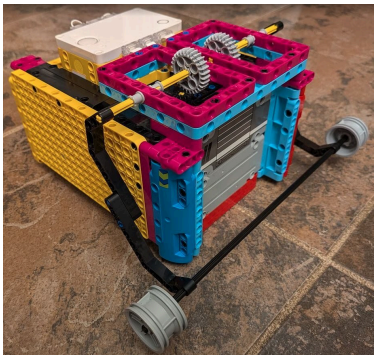
- It's fine to read your words. Look up at the judges between sentences.
- Talk slower than feels normal, and loud enough for the back of the room.
- Hold up the robot, or point at the picture, when your part mentions it.
- When you finish, say the next person's name and step back.
- After the 5 minutes, the judges ask questions. **Anyone** can answer.

#	Rubric	Speaker	Time	Judges' handout
1	Identify	Bowen	1:30	Pages 1–2
2	Identify	Lina	0:25	Page 3
3	Design	Miriam	0:30	Page 4
4	Design	James	0:30	Page 4
5	Create	Jennie	0:30	Page 5
6	Create	Bowen	0:15	Page 5, hold up the attachment
7	Iterate	Neala	0:45	Page 6, hold up the keystone species
8	Communicate	Nolan	0:45	Page 7
9	Communicate	Mira	0:10	Page 7 (everyone stands together)
			≈ 5:15	

Judges' handout, page 1 (top)

Rookie team · 2026 FLL Challenge: Bioglow · Robot: Rosie Rover · Coded in Pybricks (Blocks to Python)





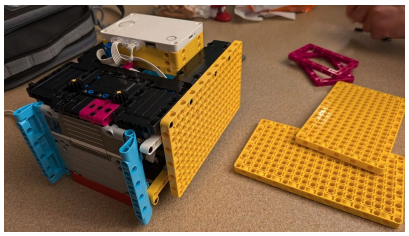
Rubric	Who's talking	What we'll show
Identify	**Bowen**, **Lina**	Resources used, robot base, Pybricks + remote, mission strategy
Design	**Miriam**, **James**	Mission 1 Drone Survey plan, motor-driven arm (Mission 2 Exploding Seeds, Mission 7 Humongous Fungus)
Create	**Jennie**	Mission 13 Keystone Species code and attachment
Iterate	**Neala**	How testing changed Mission 13 Keystone Species
Communicate	**Nolan**, **Mira**	Lessons learned

1. IDENTIFY: Resources & the Robot Base (Bowen, about 1 min 30 sec)

Rubric: "explored building and coding resources, and sought guidance as needed."

On the judges' handout:

- **As a rookie team, we started by learning from veteran teams.** The YouTube video *"5 Simple Tips to Build the Ultimate FLL Robot"* recommends building one strong **base robot** first, then small attachments for each mission.
- **Base:** built following the video's tips, then finished with an original design for the outside panels. It has 2 drive motors, **2 attachment motors on top**, and 2 color sensors. We made a Pybricks template program for this setup.

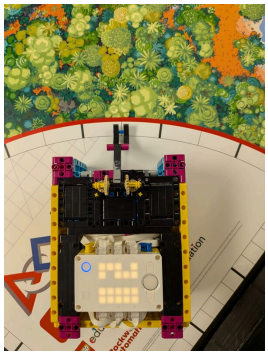




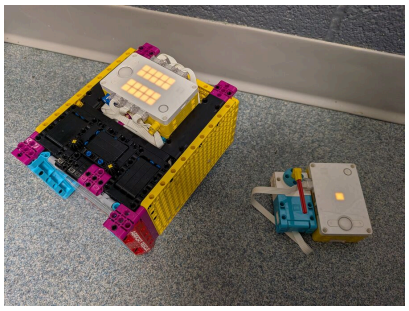
- **Pybricks instead of the regular SPIKE app:** missions start in Pybricks Blocks, then move to Python. Each mission is its own file, and all the code is on GitHub (https://github.com/Rosie-Rover-FLL/bioglow_2026).



- **Mission menu on the hub:** press left/right to pick a mission number, center to run it.



- **Custom remote control:** a second hub. Tilt it to drive, a knob sets the speed, and holding a button moves the arm. Used for driving practice and testing attachments. (live demo)



🎤 Bowen says:

> Hi, my name is Bowen Fisher. We are Rosie Robotics, and I am here to present our robot, the Rosie Rover.
>
> This is our first year as a FIRST LEGO League team, so before starting any designs, we asked for advice from teams that have done this in the past.
>
> We found videos of ideas and robot designs. We found a video with a link to build a robot base, and we used it to make most of the Rosie Rover we have today. *[point to the top gears]* These are called **dog ears**. They run our entire attachment system.
>
> Instead of the SPIKE Prime app, we use **Pybricks**. We start with Blocks, then move that code to **Python**. We wrote a main mission program that can run all of our missions: the left and right buttons pick which mission, and the center button runs it.
>
> We also built a **remote control** out of a second hub. We use it **only for testing**. *[quick demo, about 5 seconds]*
>
> Next, Lina will present how we picked our missions.

Bowen's notes (don't read out loud):

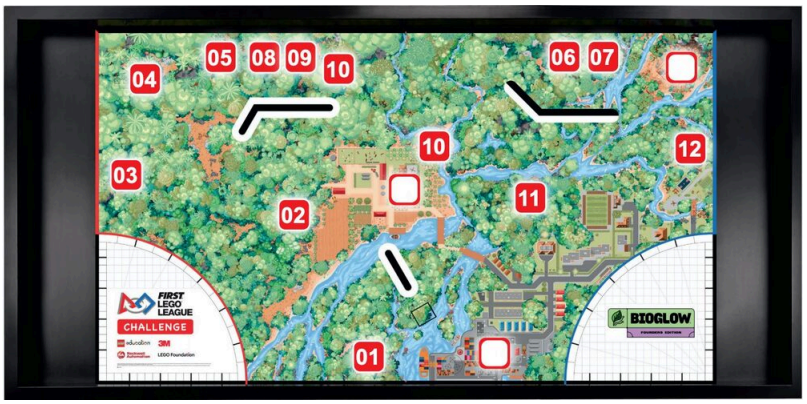
- If a judge asks "why Pybricks?": we can write Python, every mission lives in its own file, the code is all saved on GitHub, and we could build a mission menu and a remote.
- The remote is only for practice. Remote controls aren't allowed in matches.

2. IDENTIFY: Mission Strategy (Lina, about 25 sec)

Rubric: "determined which missions to attempt" (mission strategy).

On the judges' handout:

Missions chosen for **reliability** and the **easiest points**: close to a launch area, with simple moves.



Mission	Goal pts
Equipment inspection	20
01 Drone Survey	20
02 Exploding Seeds	20
03 Flip the Rock	30
06 Leafcutter Frenzy (also scores 11 Window to the Past)	50
07 Humongous Fungus	20
10 Fragile Microhabitats (don't touch it!)	20
11 Window to the Past	20
13 Keystone Species	30
15 Biocentric Architecture	10
Precision tokens	50
Goal	**290**

🎤 Lina says:

> There are 15 missions, and nobody can do them all. As a rookie team, we picked missions we can do **reliably**, and the **easiest points**: missions close to a launch area, with simple moves. The map and the table show the missions we chose. Our goal is **290 points**.
>
> Now Miriam and James will tell you how we designed our attachments.

3–4. DESIGN: Mission 1 Drone Survey and the Arm (Miriam and James)

Rubric: "all team members contributed ideas" and "building and coding skills in all team members."

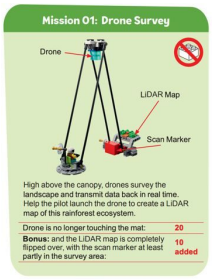
On the judges' handout:

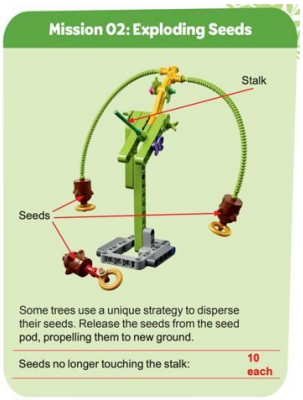
Mission 1 Drone Survey (Miriam): the robot drives backward fast to knock the drone off the mat, then drives straight home.

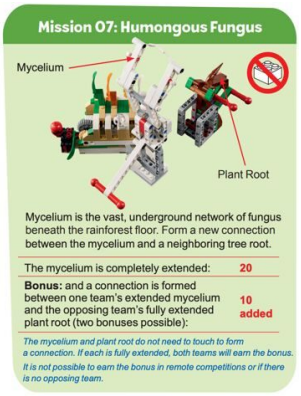
The arm, Mission 2 Exploding Seeds (James): both top motors turn the arm together. The first version was too weak, so the speed went way up. The arm lowers again on the way home, ready for the next run. We might even be able to use the same arm for **Mission 7 Humongous Fungus**.











🎤 Miriam says (about 30 sec):

> I worked on **Mission 1, Drone Survey**. Our plan was to use speed: the robot drives **backward fast** and knocks the drone off the mat. Then it drives straight home. *[CHECK: one sentence about the part of the robot that hits the drone, e.g. "We added ___ to the back so ___."]* I helped build it and code it, and we tested it a lot.
>
> I decided to try to accomplish the drone survey (mission 1). I built an attachment, but my first version did not work but when tinkered with the attachment to make it better my second version worked. Here is my attachment. The robot is turned so it is facing backwards and against the wall then it drives forward to push the red thing forward so the drone is in the air. I built the attachment,

> tested the robot, and measured how far it should go forward and backward. I tested and it worked
>
> I decided to try to accomplish the drone survey (mission 1). I built an attachment, but my first version did not work but when tinkered with the attachment to make it better my second version worked. Here is my attachment. The robot is turned so it is facing backwards and against the wall then it drives forward to push the red thing forward so the drone is in the air. I built the attachment, tested the robot, and measured how far it should go forward and backward. I tested and it worked 3 times in a row.
>
> 3 times in a row.

> Now James will tell you about our arm.

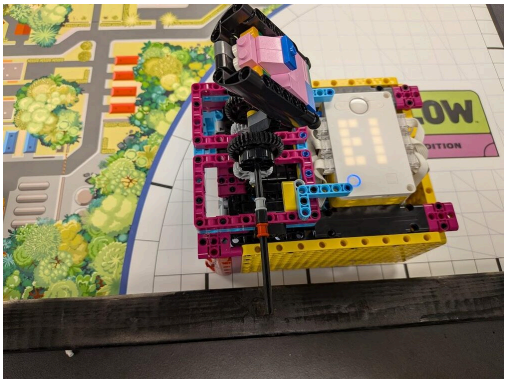
🎤 James says (about 30 sec):

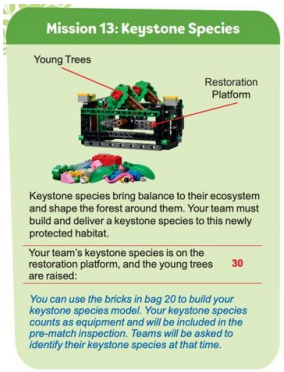
> I worked on **the arm**. Both top motors turn it together. For **Mission 2, Exploding Seeds**, the robot drives up and lifts the arm to knock the seeds off. At first the arm was **too weak**, so we made it go much faster, and then it worked. We might even be able to use the same arm for **Mission 7, Humongous Fungus**.
>
> Now Jennie will show you our code for Mission 13, Keystone Species.

5–6. CREATE: Mission 13 Keystone Species (Jennie, then Bowen)

Rubric: "clear explanation of innovative attachments and their purpose" and "clear explanation of innovative code and/or sensor use."

On the judges' handout:

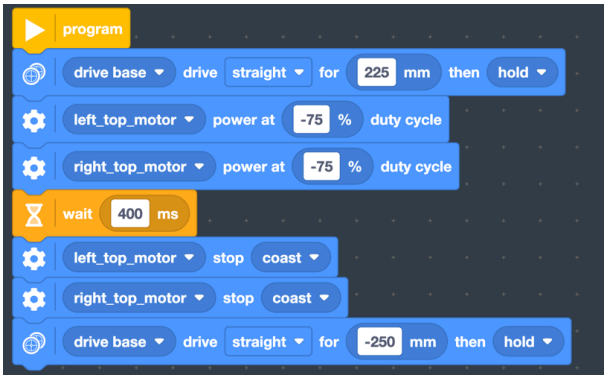




Keystone species: a poison dart tree frog. It eats huge numbers of insects and is food for other animals, so it holds the food web together.

Attachment: connects to the top motors with gears, and throws the keystone species onto the restoration platform.

Code (Jennie): first written in Pybricks Blocks. Drive to the platform, run both top motors at 75% power for 0.4 seconds, then back up.



🎤 Jennie says (about 30 sec):

> I wrote the code for **Mission 13, Keystone Species**. Our keystone species is a **poison dart tree frog**. It eats huge numbers of insects and is food for other animals, so it holds the food web together. We have to get our frog onto the restoration platform and raise the young trees. I wrote it in Pybricks Blocks, and it's short. The robot drives forward 225 millimeters. Then both top motors run at 75% power for less than half a second to **throw** our species. Then the robot backs up.
>
> Bowen will show you the attachment.

🎤 Bowen says (about 15 sec):

> This is the Mission 13 attachment. *[hold it up]* The gears connect it to the top motors, and it throws our keystone species onto the platform. *[CHECK: one more sentence about what's special about it]*

> Neala will tell you how testing changed it.

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7. ITERATE: How Testing Changed Mission 13 Keystone Species (Neala, about 45 sec)

Rubric: "repeated testing" and "improvements based on testing."

On the judges' handout:

Repeated test runs showed two problems:

Problem found in testing	Fix
Not enough power to raise the young trees	Added **weight** to the keystone species, started the attachment **farther back** for more momentum, and raised the throw from 75% to **100% power**
The species sometimes missed the center of the platform	Drive out **slower** (300 mm/s) to stop wheel slip, but still drive back **fast** (600 mm/s)



These edits were made directly in the Python code (changes marked ◀):

```
drive_base.settings(straight_speed=300)   # ◀ NEW: slow drive out, no wheel
slip
drive_base.straight(225)
left_top_motor.dc(-100)                   # ◀ was -75: full-power throw
right_top_motor.dc(-100)                  # ◀ was -75
wait(400)
left_top_motor.stop()
right_top_motor.stop()
drive_base.settings(straight_speed=600)   # ◀ NEW: fast drive home
drive_base.straight(-250)
```

🎤 Neala says:

> We tested Mission 13 **over and over**, and testing showed two problems.
>
> **First**, there wasn't enough power to raise the young trees. So we made our keystone species **heavier** *[hold it up]*. We also start the attachment **farther back**, so it has more momentum when it hits the target.

> **Second**, the species sometimes missed the center of the platform. The wheels were slipping, so we made the robot drive out **slower**. It still drives back **fast** to save time.
>
> We made those changes right in the Python code. The arrows on the handout show what we changed. Now Mission 13 works almost every time.
>
> Nolan will tell you what we learned.

8–9. COMMUNICATE: Lessons Learned (Nolan, then Mira)

Rubric: "detailed explanation of process and lessons learned" and "team clearly shows pride or enthusiasm for their work."

On the judges' handout:

1. **A robot doesn't do the same thing every time.** The battery, where the robot is placed, and wheel slip all matter. The fixes: stall-detect code (if the robot gets stuck, it stops and moves on), and gentler starts so the wheels don't slip.
2. **Precision tokens are worth protecting.** Each touch of the robot outside home costs a token. Sometimes it's smarter to let a mission fail.
3. **Practice runs taught us the scoring.**

	Score
First full practice run (Oct 11)	180
...many more runs not shown...	...
Practice run (Nov 1)	230
Practice run (Nov 8)	220
Current goal	**290**

The score sheet used at every practice run:

Mission	Got	Goal	Max
Equipment inspection All equipment in one launch area, under 12 in.		20	20
01 Drone Survey Drone off mat 20 · LiDAR bonus 10		20	30
02 Exploding Seeds 10 per seed off the stalk (3)		20	30
03 Flip the Rock Flag down 20 · rock returned 10		30	30
04 Lucky Leaves One leaf 10 · second leaf bonus 20		0	30
05 Reaching Roots Partial 10 or complete 20		0	20
06 Leafcutter Frenzy 10 per fragment + ant in the nest		50	50
07 Humongous Fungus Mycelium extended 20 · connection bonus 10		20	30
08 Tangled Vine touching the mat		0	30
09 Research Platform Platform 10 · camera trap 10 · seed 10		0	30
10 Fragile Microhabitats Spider 10 and Snail 10 left in place		20	20
11 Window to the Past Root cover down, touching the mat		20	20
12 Forest Elder Cane raised 20 · tie around post 10		0	30
13 Keystone Species Species on platform, young trees raised		30	30
14 Seeds of Renewal 5 per seed in station · +5 touching mat		0	40
15 Biocentric Architecture Canopy 10 Skylight 10 Hatch 10 Dock bonus 10		10	40
Precision tokens Points for tokens left at the end		50	50
Total	**0**	**290**	**530**

Proud of Rosie Rover! As a brand-new team, we used **Discovery, Innovation, Impact, Inclusion, Teamwork, and Fun** to build it together.

Thank you, judges!

🎤 Nolan says (about 45 sec):

> We learned three big things.
>
> **One:** a robot doesn't do the same thing every time. The battery, where you place the robot, and wheel slip all change things.
>
> **Two:** precision tokens are worth protecting. Each time we touch the robot outside home, it costs us a token. So sometimes it's smarter to let a mission fail.
>
> **Three:** practice runs taught us the scoring. We fill out this score sheet at every practice. Our first full run on October 11th scored **180**. On November 1st we got **230**, and last week **220**, which shows how much runs can change. Our goal is **290**.
>
> Mira will finish for us.

🎤 Mira says (about 10 sec):

> We're really proud of Rosie Rover! As a brand-new team, we used **discovery, innovation, impact, inclusion, teamwork,** and **fun** to build it together. Thank you, judges!

(Everyone smiles. Then we're ready for questions!)

Practice questions judges might ask

Practice these out loud. Anyone can answer.

- What was the hardest part of building the robot? How did you fix it?
- Who did the coding? Did everyone code? *(Everyone should be able to say one thing they coded.)*
- Do you use any sensors? *(We have two color sensors. CHECK: what we use them for)*
- How do you make the robot start in the same place every time?
- What would you change if you started over?
- How did you decide who works on which mission?
- What's your favorite mission and why?
- What is your keystone species, and why is it a keystone species? *(The referee also asks this at the pre-match inspection.)*